

AP CSP | Chapter 8 Answer Key

Diagnostic | Original practice rationales

Use after completing the set. Each item is one practice count; no partial credit for selecting only one required answer in a two-answer item. These counts do not estimate an official AP score. The online review includes a Chinese explanation and a link to the relevant lesson.

Q01 | CRD | Answer: B

Purpose describes the problem or need the innovation addresses. Sharing surplus food is the purpose; displaying, reading input and storing data are functions.

Review: L02 · Computing Innovations, Input & Output

Q02 | CRD | Answer: A, C

Observe the intended users, identify the barriers, revise and retest. Extra features and the developers' earlier success do not resolve the reported usability problem.

Review: L04 · Collaborate, Test & Improve

Q03 | DAT | Answer: D

Four bits provide 16 patterns, which is too few. Five bits provide 32 patterns and are sufficient; six or nineteen bits are not minimal.

Review: L01 · Bits & Binary

Q04 | DAT | Answer: A

Lossless compression allows exact recovery. A similar-looking graph, rounded values or group averages can hide irreversible changes to the original readings.

Review: L06 · Compression & Trade-offs

Q05 | AAP | Answer: C

The assignment to b copies 8. Updating a to 11 leaves b at 8, so the displayed sum is 19.

Review: L03 · Algorithms, Variables & Sequence

Q06 | AAP | Answer: B

A list lets one algorithm process a variable number of related values. It does not guarantee accurate data, constant-time searching or a missing-data policy.

Review: L17 · Lists & Data Abstraction

Q07 | CSN | Answer: D

A-C-D remains connected when A-B fails. Redundant paths can preserve communication; encryption does not replace a route.

Review: L25 · Redundancy & Fault Tolerance

Q08 | CSN | Answer: B

Schedule the 6- and 2-second tasks on one processor and the 7-second task on the other. The finish time is 8 seconds; indivisible tasks prevent reaching 7.5 seconds.

Review: L26 · Parallel & Distributed Computing

Q09 | IOC | Answer: A

Shared kiosks address some device and connectivity barriers. They do not guarantee universal access or participation, and residents need not own a smartphone to use them.

Review: L27 · Impacts & the Digital Divide

Q10 | IOC | Answer: C

Improve the relevant data coverage and evaluate the revised system across groups. More data from the same group or hiding group labels does not establish that the bias is reduced.

Review: L28 · Computing Bias & Crowdsourcing

AP CSP | Chapter 8 Answer Key

Algorithms | Original practice rationales

Use after completing the set. Each item is one practice count; no partial credit for selecting only one required answer in a two-answer item. These counts do not estimate an official AP score. The online review includes a Chinese explanation and a link to the relevant lesson.

Q11 | AAP | Answer: B

Evaluate MOD and division before addition: $23 \text{ MOD } 6 = 5$ and $15 / 4 = 3.75$. Their sum is 8.75; division is not truncated to an integer.

Review: L09 · Expressions & Strings

Q12 | AAP | Answer: D

saved preserves the original left value. After left receives the original right value, assigning saved to right completes the swap.

Review: L03 · Algorithms, Variables & Sequence

Q13 | AAP | Answer: A

Negating the inclusive inside-range condition gives exactly the outside range. AND cannot join the two outside cases, and the endpoints must remain inside.

Review: L10 · Boolean Logic

Q14 | AAP | Answer: C

The loop adds 2, 4, 6 and 8, producing 20. value is increased only after its current value is added.

Review: L13 · Counted Loops

Q15 | AAP | Answer: B

Four doublings change x from 1 to 2, 4, 8 and 16. At 16 the stopping condition is true, so steps is 4.

Review: L14 · Conditional Loops

Q16 | AAP | Answer: D

Insertion gives [4, 6, 8, 12]. Removing the current third item removes 8; appending the first item then gives [4, 6, 12, 4].

Review: L17 · Lists & Data Abstraction

Q17 | AAP | Answer: A

The first call returns 11. The second call receives 11 and 1, so it returns $2 \times 11 + 1 = 23$. Each call binds its own arguments to a and b.

Review: L19 · Procedures, Parameters & Libraries

Q18 | AAP | Answer: C

The middle item is 15. Searching the higher half next checks 28, then its lower remaining range contains only 21.

Review: L20 · Searching & Comparing Algorithms

Q19 | AAP | Answer: B

Both formulas are polynomial, so both fit the AP CSP reasonable-time classification. n^2 is not exponential, and the classification does not promise a fixed elapsed-time limit.

Review: L22 · Efficiency, Limits & Chapter Project

Q20 | AAP | Answer: D

The forward square is blocked, so the robot turns east and moves to (3, 3). The last two commands turn it north and move it to (2, 3), still facing north.

Review: L21 · Robot Algorithms & Problem Solving

AP CSP | Chapter 8 Answer Key

Mixed | Original practice rationales

Use after completing the set. Each item is one practice count; no partial credit for selecting only one required answer in a two-answer item. These counts do not estimate an official AP score. The online review includes a Chinese explanation and a link to the relevant lesson.

Q21 | CRD | Answer: C

Completed arrangements and repairs are directly related to the stated purpose. Code length, color count and alphabetical order do not measure successful repairs.

Review: L02 · Computing Innovations, Input & Output

Q22 | CRD | Answer: B, D

Broaden the study to relevant devices, times and accessibility needs, then retest changes. More evidence from the original narrow group cannot establish usability for every intended user.

Review: L04 · Collaborate, Test & Improve

Q23 | DAT | Answer: D

There are 48 pixels, each requiring 3 bits, so the payload is 144 bits. That is 18 bytes, not 18 bits; headers and metadata are excluded.

Review: L05 · Sampling & Digital Representation

Q24 | DAT | Answer: B

Method B meets the exact-recovery requirement. Method A is smaller but discards information; compression does not have to be lossy.

Review: L06 · Compression & Trade-offs

Q25 | DAT | Answer: A, D

The full mean is $160 / 5 = 32$ and the median is 18. A labeled sensitivity analysis can compare results with and without 95, but deletion alone does not prove a sample is unbiased.

Review: L08 · Data Analysis, Visualisation & Evidence

Q26 | DAT | Answer: C

Metadata reveals that the values use different units. Convert to one unit before aggregation; binary encoding does not make Celsius and Fahrenheit numerically interchangeable.

Review: L07 · Metadata, Data Quality & Cleaning

Q27 | AAP | Answer: A

The nested decision displays Silver because 80 is at least 70 but below 90. The separate final IF also runs and displays Ready.

Review: L12 · Nested Decisions & Boundary Tests

Q28 | AAP | Answer: D

Each iteration updates total before appending its current value. The running totals are 4, 6 and 11; earlier appended numbers do not change later.

Review: L18 · Traversing Lists

Q29 | AAP | Answer: B, D

The comparison is strictly greater than the limit. [4, 9] contains no value above 9, while [9, 10] does. An empty list skips the loop and returns false without an index error.

Review: L19 · Procedures, Parameters & Libraries

Q30 | AAP | Answer: B

There are three successful integers among eight equally likely results, so the expected count is $10,000 \times 3/8 = 3,750$. Random trials need not produce that exact count or a fixed quota per block.

Review: L15 · Randomness & Simulation

Q31 | AAP | Answer: C

Keep found false initially and change it to true when a match occurs, without resetting it for later items. Reversing the traversal or changing the comparison does not fix the general problem.

Review: L20 · Searching & Comparing Algorithms

Q32 | AAP | Answer: A

Undecidability rules out an algorithm that correctly resolves every instance. It does not rule out solving particular cases, and additional processing speed does not remove the proven limitation.

Review: L22 · Efficiency, Limits & Chapter Project

Q33 | AAP | Answer: D

A costs 1,000 comparisons. B costs $500 + 70 = 570$, including its one-time sorting cost, so B saves 430 comparisons under this model.

Review: L20 · Searching & Comparing Algorithms

Q34 | CSN | Answer: B

TCP uses sequence information and reliability mechanisms such as acknowledgments and retransmission. Packets need not share a route, and neither random replacement nor DNS can reconstruct missing content.

Review: L24 · Packets, Routing & Protocols

Q35 | CSN | Answer: C

DNS resolves domain names to IP addresses. A failed lookup most directly concerns that resolution function; it does not imply a need for direct physical links or that DNS provides encryption.

Review: L23 · Systems, Networks & the Internet

Q36 | IOC | Answer: A, C

The equal-size sample gives $(90\% + 30\%)/2 = 60\%$. Population-proportion weighting gives $0.4 \times 90\% + 0.6 \times 30\% = 54\%$ under the model; weighting does not guarantee the removal of other biases.

Review: L28 · Computing Bias & Crowdsourcing

Q37 | IOC | Answer: A

Encrypt for the recipient using the recipient's authentic public key; the matching private key decrypts the message. Unrelated keys do not form a pair, and the private key must remain secret.

Review: L29 · Privacy & Security

Q38 | IOC | Answer: D

Crowdsourcing can increase scale, but useful evidence still needs validation and attention to geographic coverage. More submissions alone do not guarantee accurate or representative data.

Review: L28 · Computing Bias & Crowdsourcing

Q39 | IOC | Answer: B

The proposed sale requires permission for commercial use. Recoloring, attribution alone and online availability do not remove the stated restriction; an appropriately licensed alternative can be used under its conditions.

Review: L30 · Responsible Computing

Q40 | IOC | Answer: C

Repeated location patterns can be linked with other information to infer a home or identity. Removing names or changing formats does not guarantee anonymity.

Review: L29 · Privacy & Security